

SQL ANALYTICS REPORT

E-Commerce Data · Queries & Results

Customers · Orders · Products · Returns

26 Queries Analyzed

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Q01

How many rows are in each of the four data sheets?

```
SELECT 'customers' as table_name, COUNT(*) AS rows_count FROM customers_cleaned
UNION ALL
SELECT 'orders',COUNT(*) FROM orders_cleaned
UNION ALL
SELECT 'product',COUNT(*) FROM product_cleaned
UNION ALL
SELECT 'Returns',COUNT(*) FROM returns_cleaned;
```

Query Result

table_name	rows_count
customers	2280
orders	8472
product	25
Returns	671

Q02

Join orders to customers. Some orders will not match. How many, and what happens to your revenue total if you use an INNER JOIN instead of a LEFT JOIN?

```

SELECT
*
FROM
orders_cleaned o
INNER JOIN
customers_cleaned c ON o.customer_id = c.customer_id;
SELECT
COUNT(*) AS unmatched_orders
FROM
orders_cleaned o
LEFT JOIN
customers_cleaned c ON o.customer_id = c.customer_id
WHERE
c.customer_id IS NULL;

```

Query Results

unmatched_orders	gross_revenue	net_revenue	gross_revenue	net_revenue
467	22761130	21048578.05	23986111	22198505.05

Q03

After all cleaning, what is total net revenue and how many valid orders does it cover?

```

SELECT
ROUND(SUM(o.net_revenue), 2) AS net_revenue,
COUNT(*) AS valid_order
FROM
orders_cleaned o
INNER JOIN
customers_cleaned c ON o.customer_id = c.customer_id;

```

Query Result

net_revenue	valid_order
21048578.05	8005

Q04

How much order value was lost to cancellations, in rupees and as a percentage of orders?

```

SELECT COUNT(r.order_id) AS return_order,COUNT(o.order_id) AS total_order FROM
orders_cleaned AS o
LEFT JOIN returns_cleaned AS r
ON o.order_id = r.order_id;
SELECT Round(SUM(o.net_revenue),2) AS return_order_in_rupee FROM orders_cleaned AS o
LEFT JOIN returns_cleaned AS r
ON o.order_id = r.order_id
WHERE r.return_id IS NOT NULL;
WITH sum_lost AS(
SELECT Round(SUM(o.net_revenue),2) AS return_order_in_rupee FROM orders_cleaned AS o
LEFT JOIN returns_cleaned AS r
ON o.order_id = r.order_id
WHERE r.return_id IS NOT NULL
),
pct AS(
SELECT Round(SUM(o.net_revenue),2) AS return_order_in_pct FROM orders_cleaned AS o
LEFT JOIN returns_cleaned AS r
ON o.order_id = r.order_id
)
SELECT Round((return_order_in_rupee*100/return_order_in_pct),2) AS return_rate FROM
sum_lost,pct;

```

Query Results

return_order	total_order	return_order_in_rupee	return_rate
651	8472	1469932	6.62

Q05

Net revenue by category, highest to lowest.

```
SELECT
pr.category, ROUND(SUM(oc.net_revenue), 2) AS net_revenue
FROM
orders_cleaned oc
INNER JOIN
product_cleaned AS pr ON oc.product_id = pr.product_id
GROUP BY pr.category
ORDER BY net_revenue DESC;
```

Query Result

category	net_revenue
Electronics	12036967.15
Apparel	4840136.65
Fitness	3401260.2
Home	1920141.05

Q06

Top 5 products by net revenue.

```
SELECT
pr.product_name, ROUND(SUM(oc.net_revenue), 2) AS net_revenue
FROM
orders_cleaned oc
INNER JOIN
product_cleaned AS pr ON oc.product_id = pr.product_id
GROUP BY pr.product_name
ORDER BY net_revenue DESC
```

Query Result

product_name	net_revenue
Wireless Earbuds Pro	2942833.95
Smartwatch Lite	2763439.35
Over-Ear Headphones	1956858.55
Running Shoes	1802573.6
Smart Fitness Band	1444144.05

Q07

Top 5 cities by net revenue.

```
SELECT  
pr.city, ROUND(SUM(oc.net_revenue), 2) AS net_revenue  
FROM  
orders_cleaned oc  
INNER JOIN  
customers_cleaned AS pr ON oc.customer_id = pr.customer_id  
GROUP BY pr.city  
ORDER BY net_revenue DESC  
LIMIT 5;
```

Query Result

city	net_revenue
Mumbai	4136341.6
Bengaluru	3338801.85
Delhi	3073986.4
Chennai	1895435.75
Pune	1745461

Q08

Net revenue by channel.

```
SELECT  
channel, ROUND(SUM(net_revenue), 2) AS net_revenue  
FROM  
orders_cleaned  
GROUP BY channel;
```

Query Result

channel	net_revenue
Website	7359592.5
Mobile App	11003350.7
Marketplace	3835561.85

Q09

Order count by payment method.

```
SELECT  
payment_method, COUNT(*) AS payment_count  
FROM  
orders_cleaned  
GROUP BY payment_method;
```

Query Result

payment_method	payment_count
UPI	3319
Wallet	499
Net Banking	491
Cash on Delivery	2055
Credit Card	1221
Debit Card	887

Q10

Build a monthly revenue series. Which month was the peak and which the lowest?
Explain the lowest one before you flag it as a problem

```

SELECT
MONTHNAME(order_date) AS Month_name,
ROUND(SUM(net_revenue), 2) AS Total_revenue,
COUNT(order_id) AS order_count,
SUM(CASE
WHEN order_status = 'Delivered' THEN 0
ELSE 1
END) AS order_cancel
FROM
orders_cleaned
GROUP BY MONTHNAME(order_date)
ORDER BY Total_revenue DESC;

```

Query Result

Month_name	Total_revenue	order_count	order_cancel
October	3145881.75	1163	171
November	2978178.85	1175	152
August	2049808.05	781	96
September	1903689.6	726	91
June	1838424.35	673	79
May	1732709.5	642	77
July	1670877.6	683	96
December	1624114.5	630	70
April	1518047.55	570	78
March	1313253.85	532	60
February	1227464.4	465	62
January	1196055.05	432	69

Q11

Compare full-year 2024 revenue to 2025. What is the year-on-year growth rate?

```

WITH GRWOTH_AND_REVENUE AS(
SELECT
YEAR(order_date) AS years,
ROUND(SUM(net_revenue)) AS total_sales
FROM
orders_cleaned
GROUP BY years
)
SELECT
years,
total_sales,
ROUND((total_sales - LAG(total_sales) OVER(ORDER BY years))
/ LAG(total_sales) OVER(ORDER BY years) * 100,2) AS GROWTH_RATE
FROM GRWOTH_AND_REVENUE
ORDER BY years;

```

Query Result

years	total_sales	GROWTH_RATE
2024	7500502	NULL
2025	14698003	95.96

Q12

What share of annual revenue falls in October and November? Is that a real festive effect or just noise?

```

WITH share as( SELECT
SUM(net_revenue) AS oct_nov_revenue
FROM orders_cleaned
WHERE MONTH(order_date) IN (10, 11)
),
net_Rev as(
SELECT SUM(net_revenue) as total_rev FROM orders_cleaned
)
SELECT round((oct_nov_revenue/total_rev)*100,2) as total_share_inpercent FROM share
CROSS JOIN net_Rev;

```

Query Result

total_share_inpercent
27.59

Q13

Rank each month within its year by revenue

```

WITH mom AS(
SELECT
year(order_date) AS years,
monthname(order_date) AS months_name,
round(SUM(net_revenue)) AS net_sales
FROM orders_cleaned
GROUP BY years, months_name
ORDER BY net_sales DESC
)
SELECT
years,
months_name,
net_sales,
RANK() OVER(PARTITION BY years ORDER BY net_sales DESC) as sales_rank

```

Query Results

years	months_name	net_sales	sales_rank
2024	November	1297802	1
2024	December	1123477	2
2024	October	1102018	3
2024	August	752391	4
2024	September	685885	5
2024	July	676922	6
2024	June	583789	7
2024	May	410108	8
2024	April	348115	9
2024	March	269162	10
2024	February	181824	11
2024	January	69009	12
2025	October	2043864	1

years	months_name	net_sales	sales_rank
2025	October	2043864	1
2025	November	1680376	2
2025	May	1322602	3
2025	August	1297417	4
2025	June	1254635	5
2025	September	1217805	6
2025	April	1169933	7
2025	January	1127046	8
2025	February	1045640	9
2025	March	1044091	10
2025	July	993956	11
2025	December	500637	12

Q14

Monthly net revenue by product category

```

FROM
mom;
SELECT
YEAR(ord.order_date) AS years,
MONTHNAME(ord.order_date) AS months,
ROUND(SUM(ord.net_revenue)) AS net_sale,
pr.category
FROM
orders_cleaned AS ord
INNER JOIN

```

Query Results

years	months	net_sale	category
2025	October	1133741	Electronics
2025	November	931521	Electronics
2025	June	752924	Electronics
2025	May	729706	Electronics
2025	August	719691	Electronics
2024	November	697593	Electronics
2025	April	693192	Electronics
2025	January	646060	Electronics
2025	September	611412	Electronics
2024	December	588437	Electronics
2024	October	578785	Electronics
2025	July	545394	Electronics
2025	February	544411	Electronics

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years	months	net_sale	category
2024	June	49602	Home
2024	April	49440	Fitness
2024	January	44648	Electronics
2024	March	44035	Fitness
2024	May	35338	Home
2024	March	28182	Home
2024	April	28046	Home
2024	February	22252	Fitness
2024	January	15090	Home
2024	February	9994	Home
2024	January	6125	Fitness
2024	January	3146	Apparel

Q15

Net revenue by category and sub-category

```

product_cleaned AS pr ON ord.product_id = pr.product_id
GROUP BY months , category , years
ORDER BY net_sale DESC;
SELECT pr.category,pr.sub_category,round(SUM(net_revenue)) as net_sale FROM
product_cleaned pr
INNER JOIN orders_cleaned ord

```

Query Result

category	sub_category	net_sale
Electronics	Audio	5728478
Electronics	Wearables	4207583
Apparel	Footwear	2516493
Electronics	Accessories	2100906
Fitness	Equipment	1867680
Fitness	Nutrition	1533581
Apparel	Menswear	1288414
Home	Kitchen	1061654
Apparel	Womenswear	1035229
Home	Furnishing	858487

Q16

Calculate total gross profit and overall margin percentage.

```

SELECT
SUM(gross_profit) AS gross_profit
FROM
product_cleaned;

```

Query Result

gross_profit
28005

Q17

Overall margin percentage (revenue-based formula)

```

SELECT
FLOOR((SUM(od.net_revenue) - SUM(cost_price)) / SUM(od.net_revenue) * 100) AS
margin_percentage
FROM
orders_cleaned AS od
INNER JOIN
product_cleaned AS pr ON od.product_id = pr.product_id;

```

Query Result

margin_percentage
64

Q18

Margin percentage by category, highest to lowest.

```

SELECT
pr.category,
FLOOR((SUM(od.net_revenue) - SUM(cost_price)) / SUM(od.net_revenue) * 100) AS
margin_percentage,
ROUND(SUM(od.net_revenue), 2) AS net_revenue
FROM
orders_cleaned AS od
INNER JOIN
product_cleaned AS pr ON od.product_id = pr.product_id
GROUP BY pr.category
ORDER BY margin_percentage DESC, net_revenue DESC;

```

Query Result

category	margin_percentage	net_revenue
Home	67	1920141.05
Apparel	66	4840136.65
Electronics	64	12036967.15
Fitness	62	3401260.2

Q19 What is the average delivery time across all orders?

```
SELECT  
ROUND(AVG(delivery_days)) AS avg_delivery_time_days  
FROM  
orders_cleaned;
```

Query Result

avg_delivery_time_days
5

Q20 Average delivery time by channel.

```
SELECT  
channel,  
ROUND(AVG(delivery_days)) AS avg_delivery_time_day  
FROM  
orders_cleaned  
GROUP BY channel;
```

Query Result

channel	avg_delivery_time_day
Website	4
Mobile App	3
Marketplace	9

Q21 Compare average delivery time in the metros against Jaipur, Kochi, Lucknow, Indore and Ahmedabad. How big is the gap?

```

WITH city as (
SELECT round(AVG(ord.delivery_days)) as ordinary_cities_delivery_date FROM
customers_cleaned as cust
INNER JOIN orders_cleaned as ord
on cust.customer_id = ord.customer_id
WHERE cust.city in ('jaipur','Kochi','Lucknow','Indore','Ahmedabad')
),
metro as (
SELECT cust.city,round(AVG(ord.delivery_days)) as metro_city_delivery_days FROM
customers_cleaned as cust
INNER JOIN orders_cleaned as ord
on cust.customer_id = ord.customer_id
WHERE cust.city not in ('jaipur','Kochi','Lucknow','Indore','Ahmedabad')
GROUP BY cust.city
)
SELECT * FROM city,metro;

```

Query Result

ordinary_cities_delivery_date	city	metro_city_delivery_days
7	Mumbai	4
7	Bengaluru	4
7	Hyderabad	4
7	Chennai	4
7	Delhi	4
7	Pune	4
7	Kolkata	4

Q22

What is the overall return rate?

```

SELECT
ROUND((COUNT(DISTINCT rt.return_id) / COUNT(DISTINCT ord.order_id ))*100,2) as return_rate
FROM
orders_cleaned AS ord
LEFT JOIN
returns_cleaned AS rt ON ord.order_id = rt.order_id;

```

Query Result

return_rate
7.69

Q23

Return rate by category. Which one is the outlier?

```

SELECT
pd.category,
ROUND((COUNT(DISTINCT rt.return_id) / COUNT(DISTINCT ord.order_id ))*100,2) as return_rate
FROM
orders_cleaned AS ord
LEFT JOIN
returns_cleaned AS rt ON ord.order_id = rt.order_id
LEFT JOIN product_cleaned as pd
on ord.product_id = pd.product_id
GROUP BY pd.category;

```

Query Result

category	return_rate
Apparel	15.58
Electronics	4.16
Fitness	6.01
Home	4.41

Q24

Drill into sub_category. Which four have the worst return rates?

```

SELECT
pd.sub_category,
ROUND((COUNT(DISTINCT rt.return_id) / COUNT(DISTINCT ord.order_id ))*100,2) as return_rate
FROM
orders_cleaned AS ord
LEFT JOIN
returns_cleaned AS rt ON ord.order_id = rt.order_id
LEFT JOIN product_cleaned as pd
on ord.product_id = pd.product_id
GROUP BY pd.sub_category
ORDER BY return_rate DESC;

```

Query Result

sub_category	return_rate
Womenswear	16.70
Menswear	15.85
Footwear	14.38
Nutrition	6.58
Furnishing	6.39
Equipment	5.60
Wearables	5.18
Audio	3.86
Accessories	3.85
Kitchen	3.40

Q25

What are the most common return reasons? Does the top reason line up with your answer to Q27?

```

SELECT
rt.return_reason, COUNT(return_id) AS total_return
FROM
orders_cleaned AS ord
LEFT JOIN
returns_cleaned AS rt ON ord.order_id = rt.order_id
GROUP BY rt.return_reason
HAVING total_return > 0
ORDER BY total_return DESC;

```

Query Result

return_reason	total_return
Size issue	211
Not as described	98
Changed mind	96
Defective product	78
Late delivery	65
Damaged in transit	58
Wrong item sent	45

Q26

Profile each channel on four measures at once: average discount, margin percentage, return rate and delivery time. One channel is bad on all four. Which, and by how much?

```

WITH channel_profile AS (
SELECT
od.channel,
ROUND(AVG(od.discount_pct), 2) AS avg_discount,
ROUND(
(SUM(od.net_revenue) - SUM(pr.cost_price * od.quantity))
/ SUM(od.net_revenue) * 100,
2
) AS margin_percentage,
ROUND(
COUNT(DISTINCT rt.order_id) * 100.0
/ COUNT(DISTINCT od.order_id),
2
) AS return_rate,
ROUND(AVG(od.delivery_days), 2) AS avg_delivery_days
FROM orders_cleaned od
INNER JOIN product_cleaned pr
ON od.product_id = pr.product_id
LEFT JOIN returns_cleaned rt
ON od.order_id = rt.order_id
GROUP BY od.channel
)
SELECT *
FROM channel_profile
ORDER BY channel;

```

Query Result

channel	avg_discount	margin_percentage	return_rate	avg_delivery_days
Marketplace	0.2	44.68	11.52	8.51
Mobile App	0.04	53.15	7.01	3.46
Website	0.04	53.1	6.35	4.08